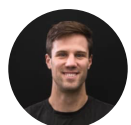


October 3, 2025

Eccentric Power – a new bar speed metric

Eccentric qualities are a key element overlooked in a lot of athletic development settings with traditional bar speed tracking technologies. Eccentric power is a new way to measure and track eccentric work in the gym



Jacob Tober
Co-founder

Bar speed tracking (often called velocity based training) has revolutionized how coaches and athletes approach strength and power development, providing real-time feedback on lifting performance with metrics like mean & propulsive velocity, peak power, and time to peak velocity.

While significant effort and research has gone into tracking concentric performance, far less has been done to quantify an athlete's ability to rapidly decelerate and control loads during the eccentric phase.

Enter eccentric power—a new metric that quantifies the watts an athlete generates (absorbs) when catching and controlling a falling load. While not appropriate for all exercises and contexts (most commonly eccentric power is useful measuring drop catch type exercises, or countermovement actions like jumping and throwing), this new metric fills a significant gap in traditional lifting analysis and provides coaches with objective data on a physical quality that has historically not been tracked objectively.

Shout out to the many coaches who helped shape this new metric through your feedback, insights and experience on the gym-floor. Special thanks to Chris C. who inspired me to write this post and actually get this metric shipped and into the app in mid-2025!

The Science Behind Eccentric Performance

Eccentric muscle actions play a fundamental role in human movement. The ability to rapidly decelerate loads (including your own bodyweight) requires exceptional eccentric strength and power.

This rapid force production mirrors the demands of countless sporting movements. In fact, eccentric capacity is arguably more critical than concentric strength for athletic performance. Every step, plant, cut, and landing an athlete takes involves eccentric loading as they absorb ground reaction forces. Every explosive concentric movement begins with an eccentric pre-stretch that stores elastic energy ready to propel the athlete forwards or vertically.

Many injuries can be traced to an athlete's lack of eccentric strength and inability to absorb eccentric loads. When deceleration capacity is lacking, the body relies on passive structures—ligaments, tendons, joint capsules, and bone—to dissipate forces that should be controlled by the muscles. This is why eccentric strength and power training has become a cornerstone of injury prevention protocols across sports.

And this is why it is important that we quantify and measure the eccentric capabilities in our athletes.

The Challenge with Eccentric Metrics

Traditional eccentric measurements face several fundamental challenges that limit their usefulness for coaches and athletes. Understanding these limitations is crucial for appreciating why a new approach is necessary.

Problem 1: Tempo Captures Volume, Not Intent

Current eccentric metrics primarily focus on duration—how long the eccentric lowering phase takes. While eccentric tempo provides useful information about control, training volume, and time under tension, it tells us nothing about the quality or intensity of the eccentric effort.

Tempo is great in a bodybuilding, rehabilitation, and general strength development contexts, but the duration of our eccentrics tells us nothing about the force and power characteristics of that eccentric.

Problem 2: The Gravity and ROM Limitation

The simple solution might be to simply measure eccentric velocity,

however this also faces a fundamental issue: mean and peak eccentric velocity are largely dictated by gravity and an athlete's range of motion rather than their actual eccentric abilities. This creates inherent bias that makes comparisons between athletes meaningless.

Consider two athletes performing drop catches:

Athlete A (6'4" with long limbs)

- Range of motion: 70cm
- Time available for gravity to accelerate the bar: Longer
- Peak eccentric velocity: -3.2 m/s
- Mean eccentric velocity: -1.9 m/s
- Traditional assessment: "Excellent eccentric speed"

Athlete B (5'6" with shorter limbs)

- Range of motion: 45cm
- Time available for gravity to accelerate the bar: Shorter
- Peak eccentric velocity: -2.1 m/s
- Mean eccentric velocity: -1.3 m/s
- Traditional assessment: "Poor eccentric speed"

Both athletes may very well possess identical braking capacities, but the taller athlete appears superior simply due to biomechanical advantage of being further away from the ground. The longer distance allows the bar more time to accelerate, creating velocity readings that are not truly comparable between athletes, and because gravity is doing most of the work to dictate velocity eccentric bar speeds are not

a very improvable metric.

Problem 3: False Progress Indicators

Take the eccentric velocity game to its logical conclusion and eccentric velocity is just too easy to cheat — just drop the bar for the maximum score.

This creates a fundamental validity problem of eccentric velocity: the measurement system cannot distinguish between athlete-controlled deceleration and external mechanical stopping.

Athletes might:

- Allow longer drop phases to inflate velocities
- Reduce their range of motion to create faster apparent stopping durations
- Fail to maintain proper control while still registering high velocity numbers

These scenarios create misleading data that doesn't reflect genuine braking capacity. High eccentric velocity tells us the barbell was moving fast, but provides no information about the athlete's ability to control that speed—the actual skill we are trying to develop and assess.

The Solution: Eccentric Power

Eccentric power, measured in watts (W) addresses these problems by measuring what matters most: the athlete's capacity to generate braking force. Rather than focusing on how fast the barbell moves

(which gravity largely determines), eccentric power quantifies how effectively the athlete can absorb and control that downward bar speed.

By measuring the rate of energy absorption during just the **active** deceleration phase, eccentric power highlights the power generated by the athlete between the moment the weight reaches its peak eccentric velocity and when they bring the barbell to a complete stop (zero velocity).

This approach captures the part of the eccentric where an athlete is working to decelerate the bar, with a score that reflects the effectiveness of the braking phase while ignoring any gravitational and biomechanical factors that skew standards eccentric velocity measurements.

How Eccentric Power is Calculated

The calculation for eccentric power isolates the true braking phase of the movement, reflecting an athlete's capacity to rapidly generate force during the lengthening phase of muscle contraction—a critical component of athletic performance that extends far beyond the weight room.

To ensure eccentric power only captures genuine deceleration ability rather than cheating or dropping the bar the calculation has multiple steps all done automatically by the Metric app.

Why This Approach Works

This methodology prevents common manipulation strategies. Athletes cannot achieve high eccentric power scores by:

- **Shortening range of motion:** The metric only measures from peak velocity onwards
- **Controlled descents:** Low peak velocities result in low kinetic energy and low power scores
- **Extended braking phases:** Longer deceleration times reduces power
- **External stopping:** Validation protocols filter out when the bar is dropped

The athlete must commit to a genuine drop of the weight followed by a hard stop to generate meaningful peak eccentric velocity.

When and How to Use Eccentric Power

Any repetition that is stopped by muscular effort will have an eccentric power value in Metric, however, knowing how to best apply it and when it provides meaningful insight is important.

Eccentric power can be measured on bilateral and unilateral exercises for both upper and lower body.

Drop Catch Training

Eccentric power is best used for exercises where athletes intentionally allow the weight to be accelerated under gravity before they then try to apply rapid braking force. This includes:

- Trapbar drop catch
- Drop lunge/catch lunges

- Drop squats from various heights
- Bench press drop-catch

Countermovement and dip exercises

For explosive movements where a countermovement is used to harness the stretch shortening cycle for greater concentric power production

- Jerk and push press exercises
- Countermovement jumps (barbell and trapbar) – measure the eccentric power on the landing for force absorption testing
- Repeated/continuous jumps – Measure the elastic ability of the athlete to rapidly absorb create elastic energy off each landing
- Rapid elastic bench press or barbell throws – measure the eccentric power before each concentric

The higher your ability to generate eccentric power will be a key factor in how well you can utilise/absorb the elastic energy of the eccentric phase to generate concentric power. Low scores on eccentric power (in both catch exercises and in countermovements is a good indicator of the need to train and develop this quality).

Programming Considerations

Load Selection Eccentric power measurements are most meaningful when using loads that challenge the athlete's braking capacity. This can be a wide range of loads anywhere from 50–110%* of the athlete's 1RM in the corresponding lift, though individual tolerance varies

significantly.

*Setup and safety for loads above 90% need to be strongly considered and tested progressively as this is an advanced application. I would recommend that the highest value will be in the 40–75% of 1RM range with a high intent and power–focus on each eccentric rep.

Volume and Recovery Drop catch exercises place substantial stress on the neuromuscular system. Limit eccentric power sessions to 3–4 sets of 1–3 quality repetitions, with extended rest periods (3–5 minutes) between attempts to ensure maximum braking capacity on each rep.

Periodization Integration Eccentric power training fits particularly well during:

- **Preparation phases:** Building robust deceleration capacity
- **Competition preparation:** Sport-specific braking demands
- **Return-to-play protocols:** Objective assessment of load tolerance

Practical Implementation

Session Structure

Warm-up Protocol Begin with extensive general warm-up followed by specific preparation:

1. Dynamic movement preparation (10–15 minutes)
2. Progressive loading to working weights
3. 2–3 submaximal drop catches for technical preparation

Assessment Protocol

An assessment session can be conducted to collect eccentric power baselines before programming these into your workouts.

1. Perform drop catches across a range of loads, working from 40% – 85% of an estimated 1RM
2. Perform 1–3 drop catches with maximum braking intent for each weight
3. Review the eccentric power chart in the Metric app and find the point that has the largest power value

Monitoring Progress

Weekly Tracking Monitor eccentric power outputs during regular training sessions to assess:

- Technical improvements in braking capacity
- Readiness for progressive loading
- Fatigue accumulation from previous sessions

The Future of Eccentric Assessment

Eccentric power represents a significant improvement in how we measure and track eccentric training in the weight room, providing coaches with objective data on a previously unmeasured athletic quality that can be assessed on any barbell lift, with minimal equipment and a time-efficient protocol. As more coaches integrate eccentric power into their assessment and training protocols, we are

excited to see the development of sport-specific normative data and refined programming strategies.

The ability to quantify braking capacity opens new avenues for:

- Injury prevention through improved deceleration training
- Return-to-play protocols with objective benchmarks
- Sport-specific preparation targeting eccentric demands
- Long-term athletic development with comprehensive force assessment

Research interest

If you are interested in conducting peer reviewed research into eccentric power, or using the Metric app in your study please do get in touch, we would love to assist however we can: jacob@metric.coach.

Conclusion

Eccentric power fills a critical gap in velocity-based training by quantifying an athlete's capacity to rapidly decelerate and control loads. This metric provides coaches with objective data to assess drop catch quality, monitor eccentric strength development, and make informed decisions about load progression and readiness.

By measuring the watts of braking force generated during the most challenging phase of deceleration, eccentric power offers unprecedented insight into a fundamental athletic quality. As coaches begin incorporating this metric into their assessment protocols, we anticipate significant advances in training specificity and athlete

development outcomes.

The integration of eccentric power into comprehensive barbell motion analysis is a big addition to objective performance measurement, providing coaches and athletes with the tools necessary to optimize both sides of the force-velocity relationship.

[← Return to blog home](#)

By lifters for lifters

Metric was built with lifters and coaches front of mind.

Focused analytics and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.



[Metric App](#)

[Available data](#)

[Support](#)



METRIC

Barbell velocity tracking

Metric for
Coaches

Metric User
Guide

Validation

& metrics

Supported
devices

Blog

Release Notes

Privacy policy

Terms &
Conditions

© Core Advantage Pty Ltd. All rights reserved.

